MATLAB Script – Set 4

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Set 4:

Given the surface charge density, $\rho_S = 2.0\,\frac{\mu C}{m^2}$, existing in the region $\rho < 1.0\ m, z = 0$ and zero elsewhere, find $\vec{E}$ at $P(\rho = 0, z = 1.0)$ and write a MATLAB program to verify your answer.

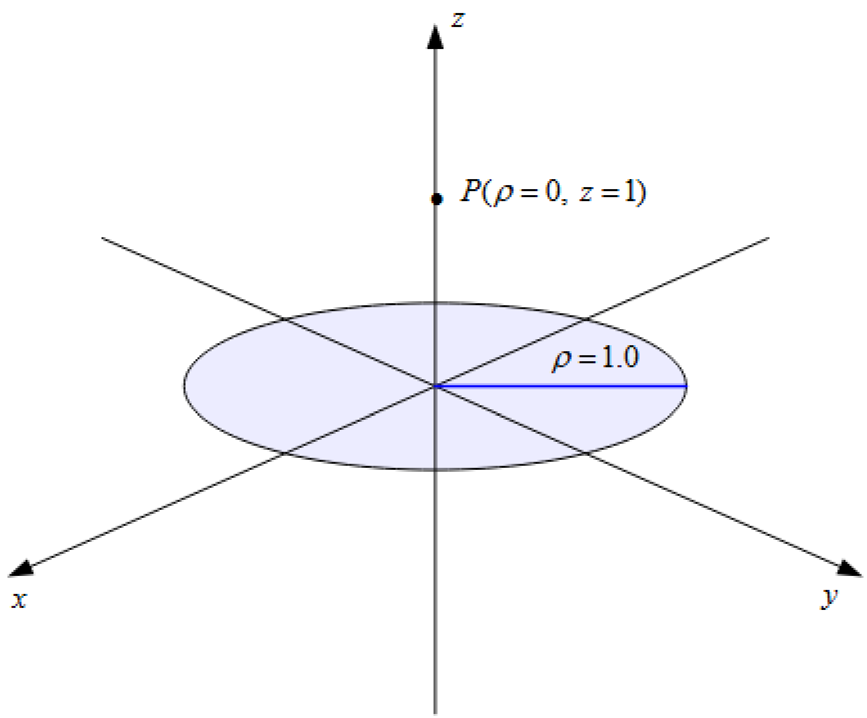

Figure 4.6 The problem of the exercise of Set 4.

Analytical Result:

The observation point, $P(0,0,1)$, is on the $z-$axis directly above the center of the disk. Thus for any surface element at position $(x,y,0)$, there exists a symmetric element at $(-x,-y,0)$. The x and y components of their electric fields cancel at $P(0,0,1)$.

$$Thus, \vec{E}\ only\ has\ a\ z-component\ and\ \vec{E} = \vec{E}_z$$

$$Given: \varepsilon_0 = 8.854 \times 10^{-12}\,\frac{C^2}{Nm^2}, \rho_S = 2.0\,\frac{\mu C}{m^2}$$

Charge element:

$$dS = \rho d\rho d\phi$$

$$dQ = \rho_S\, dS$$

$$dQ = \rho_S \rho d\rho d\phi$$

Find vector $\vec{R}$ from source to P(0, 0, 1):

$$Source\ point: P_S = (p\cos\phi, p\sin\phi, 0)$$

$$\vec{R} = P - P_S$$

$$\vec{R} = (0 - p\cos\phi)a_x - (0 - p\sin\phi)a_y + (1-0)a_z$$

$$\vec{R} = -p\cos\phi\, a_x - p\sin\phi\, a_y + a_z$$

$$|\vec{R}| = \sqrt{p^2+1}$$

Finding $\vec{E}_z$ (electric field z-component):

$$dE = \frac{dQ}{4\pi\varepsilon_0 |\vec{R}|^3}\vec{R}$$

$$dE_z = \frac{dQ}{4\pi\varepsilon_0 |\vec{R}|^3}\vec{R}_z$$

$$dE_z = \frac{\rho_S \rho d\rho d\phi}{4\pi\varepsilon_0 |\vec{R}|^3}\vec{R}_z$$

$$dE_z = \frac{\rho_S \rho d\rho d\phi}{4\pi\varepsilon_0 (\sqrt{p^2+1})^3}(1)$$

$$dE_z = \frac{\rho_S \rho d\rho d\phi}{4\pi\varepsilon_0 (p^2+1)^{\frac{3}{2}}}$$

$$\boldsymbol{Integrate}:$$

$$\vec{E}_z = \int_{\rho=0}^{\rho=1}\int_{\phi=0}^{\phi=2\pi} dE_z$$

$$\vec{E}_z = \int_{\rho=0}^{\rho=1}\int_{\phi=0}^{\phi=2\pi} \frac{\rho_S \rho}{4\pi\varepsilon_0 (p^2+1)^{\frac{3}{2}}} d\rho d\phi$$

$$\vec{E}_z = \frac{\rho_S}{4\pi\varepsilon_0}\int_{\rho=0}^{\rho=1}\int_{\phi=0}^{\phi=2\pi} \frac{\rho}{(p^2+1)^{\frac{3}{2}}} d\rho d\phi$$

$$\vec{E}_z = \frac{\rho_S}{4\pi\varepsilon_0}\int_{\rho=0}^{\rho=1} \frac{\rho}{(p^2+1)^{\frac{3}{2}}} \phi\Big|_{\phi=0}^{\phi=2\pi} d\rho$$

$$\vec{E}_z = \frac{\rho_S}{4\pi\varepsilon_0}\int_{\rho=0}^{\rho=1} \frac{\rho}{(p^2+1)^{\frac{3}{2}}}(2\pi)\,d\rho$$

$$\vec{E}_z = \frac{2\pi\rho_S}{4\pi\varepsilon_0}\int_{\rho=0}^{\rho=1} \frac{\rho}{(p^2+1)^{\frac{3}{2}}}\,d\rho$$

$$\vec{E}_z = \frac{\rho_S}{2\varepsilon_0}\int_{\rho=0}^{\rho=1} \frac{\rho}{(p^2+1)^{\frac{3}{2}}}\,d\rho$$

$\boldsymbol{U - sub}$:

$$u = p^2 + 1, du = 2pdp, thus\ pdp = \frac{du}{2}$$

$When\ p = 0, u = 1\ and\ when\ p = 1, u = 2. Thus\ new\ integration\ limits\ are\ 1 \leq u \leq 2.$

$$\vec{E}_z = \frac{\rho_S}{2\varepsilon_0}\int_{u=1}^{u=2} \frac{1}{u^{\frac{3}{2}}}\left(\frac{du}{2}\right)$$

$$\vec{E}_z = \frac{\rho_S}{2\varepsilon_0}\left(\frac{1}{2}\right)\int_{u=1}^{u=2} \frac{1}{u^{\frac{3}{2}}}\,du$$

$$\vec{E}_z = \frac{\rho_S}{4\varepsilon_0}\int_{u=1}^{u=2} \frac{1}{u^{\frac{3}{2}}}\,du$$

$$\vec{E}_z = \frac{\rho_S}{4\varepsilon_0}\int_{u=1}^{u=2} u^{-\frac{3}{2}}\,du$$

$$\vec{E}_z = \left(\frac{\rho_S}{4\varepsilon_0}\right)\frac{u^{-\frac{1}{2}}}{-\frac{1}{2}}\Bigg|_{u=1}^{u=2}$$

$$\vec{E}_z = \left(\frac{\rho_S}{4\varepsilon_0}\right)\left(\frac{(2)^{-\frac{1}{2}}}{-\frac{1}{2}} - \frac{(1)^{-\frac{1}{2}}}{-\frac{1}{2}}\right)$$

$$\vec{E}_z = \left(\frac{\rho_S}{4\varepsilon_0}\right)\left(-\frac{2}{\sqrt{2}} + 2\right)$$

$$\vec{E}_z = \left(\frac{2\rho_S}{4\varepsilon_0}\right)\left(1 - \frac{1}{\sqrt{2}}\right)$$

$$\vec{E}_z = \left(\frac{\rho_S}{2\varepsilon_0}\right)\left(1 - \frac{1}{\sqrt{2}}\right)$$

$\boldsymbol{Sub\ in\ values}$:

$$\vec{E}_z = \left(\frac{2.0 \times 10^{-6} \frac{C}{m^2}}{2\left(8.854 \times 10^{-12} \frac{C^2}{Nm^2}\right)} \right) \left(1 - \frac{1}{\sqrt{2}}\right)$$

$$\vec{E}_z \approx 33080 \frac{V}{m} \approx 3.31 \ \times 10^4 \frac{N}{C}$$

Finding $\vec{E}$ (electric field):

$$\therefore \, since \ \vec{E} = \vec{E}_z,$$

$$\vec{E} \approx 3.31 \ \times 10^4 a_z \ \frac{N}{C}$$

$$or$$

$$\vec{E} \approx 0a_x + a_y + 3.31 \ \times 10^4 a_z \ \frac{N}{C}$$

$$\therefore \, the \ total \ electric \ field \ at \ P(\rho = 0, z = 1.0) \ is \ approximately$$

$$0a_x + a_y + 3.31 \ \times 10^4 a_z \ \frac{N}{C}$$

$$which \ was \ close \ to \ the \ result \ the \ MATLAB \ code \ generated \ verifying \ our \ results.$$

$$The \ MATLAB \ code \ generated \ an \ answer \ of \approx \ 0a_x + 0a_y + 3.31 \ \times 10^4 a_z \frac{N}{C}$$

$$To \ generate \ an \ even \ closer \ result, the \ number \ of \ discretization \ steps \ for \ the \ integration \ in \ the \ MATLAB \ code \ would \ be \ increased \ (equivalently, the \ step \ size \ would \ be \ decreased), which \ would \ improve \ accuracy \ at \ the \ cost \ of \ greater \ computational \ expense.$$

MATLAB Output to Verify Answers:

```
Command Window
The electric field for the surface charge density is:
E = [0.0000  -0.0000  33080.8812] V/m
fx >>
```